Plot plan, elevations and piping isometrics

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Teaching document. Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data. Positions, spacing values, platforms, supports and the plot itself are layout assumptions for teaching. No vendor, client or contractor data. Checker and approver are left blank on purpose.

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1 Purpose, scope and basis

This document places the MEA CO2 capture unit (absorber T-101, stripper T-102, 30 wt% MEA) on a plot: the **plot plan** with every item of the equipment list and the spacing rules that set the distances, the **elevations** with the platforms, the level each field instrument is read from and the NPSH that sets the vessel heights, and five **piping isometrics** of the main lines with their line tables and bills of materials. The same data file, `layout.json`, drives the Layout & isometrics tab and the Visual plant 3D tab of the simulator, including the walking route of the outside operator.

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1.1 References

- CC-PID-101 rev A
- MEA-ME-EQL-001 rev 0
- MEA-PE-HMB-001 rev 0
- MEA-IN-INL-001 rev 0
- MEA-PE-RLB-001 rev 0
- Simulator model src/model.js (design point, noise off): K constants, HV_DEF hand valves, levels and densities

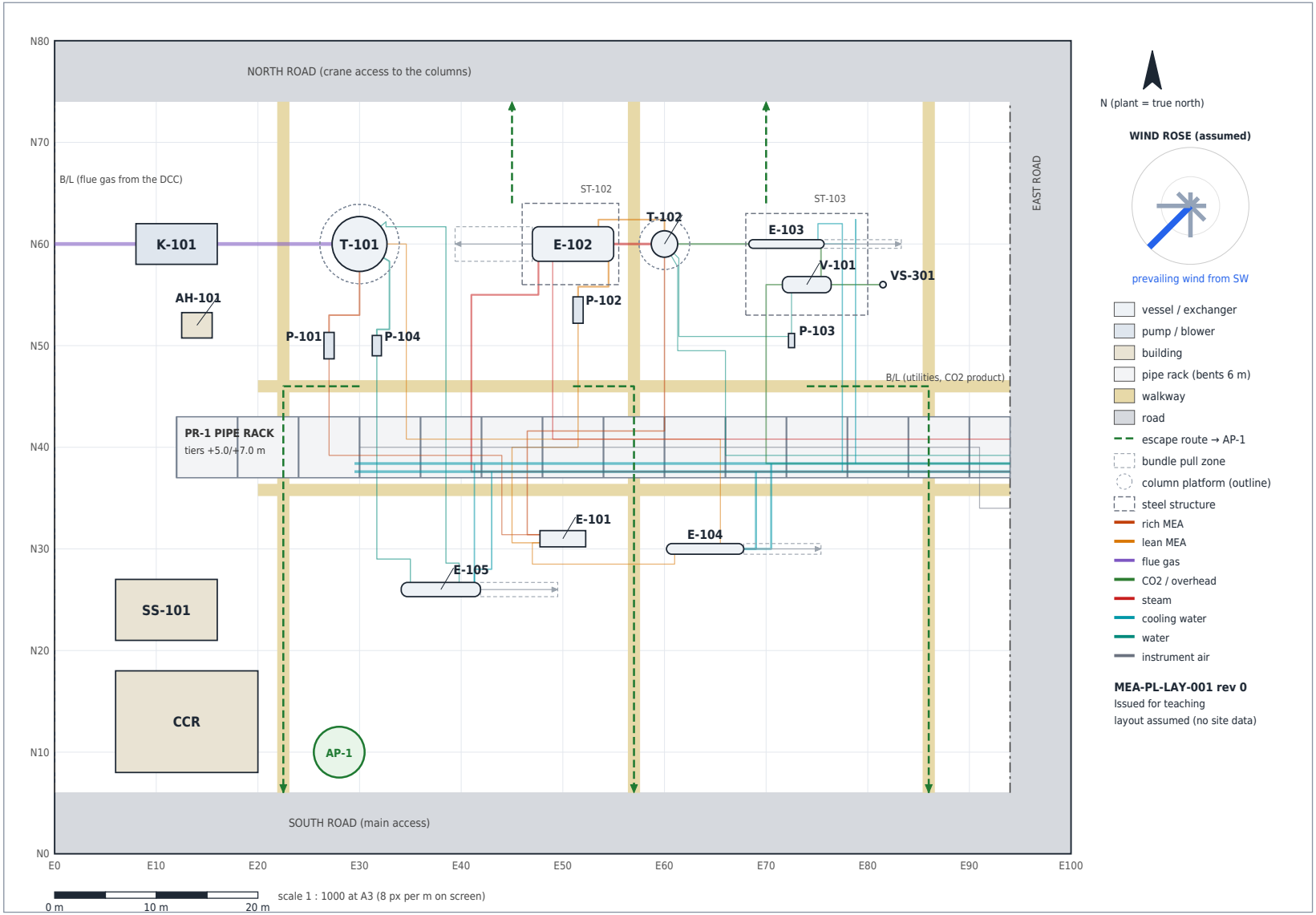
1.2 Conventions

- Units m; x east, y north, z up; grade z = 0; origin at the south-west corner of the 100 × 80 m plot. Plant north = true north (assumed).
- Elevations are centre-line (pipes), base (equipment) or floor (platforms) unless stated. The operator eye point is 1.6 m above the floor he stands on.
- Line numbers are those of CC-PID-101 rev A and the line list; numbers not on the P&ID are listed in EQ-L02. Pipe sizes in the 3-D model and the BOM are the sized DN of MEA-PE-HMB-001 rev 0 (EQ-L03).
- Spacing values are typical practice for a non-flammable aqueous amine unit, assumed for teaching; a project takes them from the owner standard and a facility siting study.
- Isometrics are not to scale (each axis compressed); the EL marks and run lengths written on them are true.

1.3 Hand valves

The 14 hand valves of the simulator (HV_DEF) are not drawn on CC-PID-101 rev A. They are placed on their lines here for the field round and flagged "added for the field round (EQ-L01)". The table in appendix A lists them.

2 Plot plan



Plot plan at scale (8 px per m in the tab). Equipment footprints from the datasheets; pipe runs in service colours; bundle pull zones dashed; escape routes in green to the assembly point AP-1. Layout assumed, no site data.

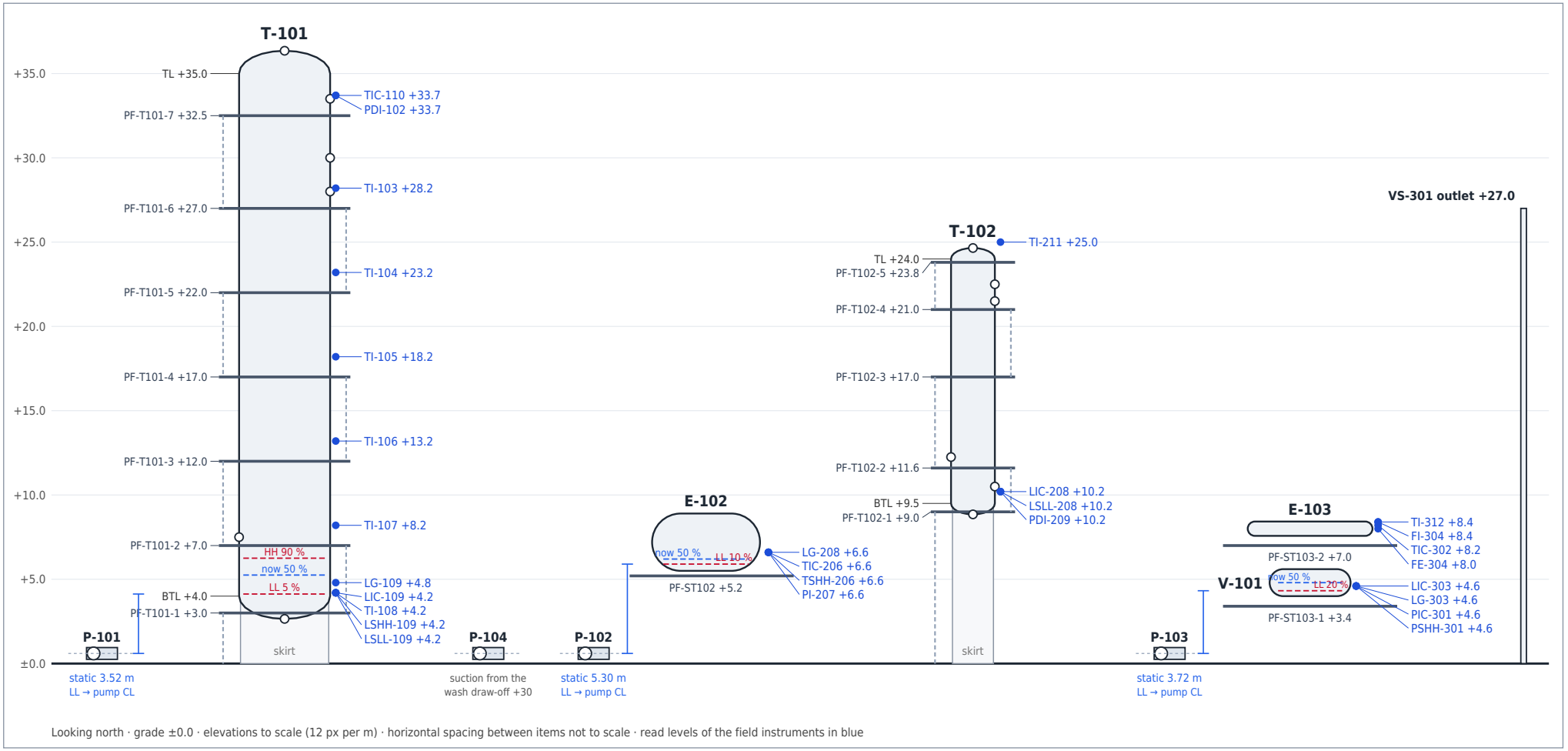
2.1 Equipment locations (the table Visual plant 3D reads)

Tag	Type	E, m	N, m	Base, m	Size, m	Description
T-101	column	30.0	60.0	+0.00	D 5.40 × H 36.4	D 5.4 m, 2 beds × 8.5 m Mellapak Plus 252Y + 3 m wash bed; skirt to 4 m (EQL)
T-102	column	60.0	60.0	+0.00	D 2.60 × H 24.6	D 2.6 m, 8 m IMTP 50 (≈ 5 stages); skirt to 9.5 m (EQL)
E-102	kettle	51.0	60.0	+5.50	D 3.40 × L 8.0	Kettle reboiler, shell D 3.4 m (K.Dr), 1582 U-tubes × 6.1 m; bottom +5.5 m (EP_EL)
V-101	drum h	74.0	56.0	+4.00	D 1.60 × L 4.8	D 1.6 m × 4.0 m T/T, 8 m³ (model K.Vdrum); bottom at 4 m (EQL)
E-103	shell tube	72.0	60.0	+7.58	D 0.85 × L 7.4	877 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 0.77 m; shell ID ≈ 0.85 m (EQL)
E-104	shell tube	64.0	30.0	+0.97	D 1.05 × L 7.6	1480 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 0.98 m; shell ID ≈ 1.05 m (EQL)
E-105	shell tube	38.0	26.0	+0.80	D 1.40 × L 7.8	2877 tubes Ø19.05 × 6.1 m, 2 passes; bundle ≈ 1.32 m; shell ID ≈ 1.40 m (EQL)
E-101	plate hx	50.0	31.0	+0.30	4.5 × 1.6 × H 3.2	530 plates × 2.0 m² (typical large plate, assumed); frame 4.5 × 1.6 × 3.2 m (assumed); nozzles on the fixed (west) frame
P-101	pump	27.0	50.0	+0.00	2.6 × 1.0 × H 1.2	487 m³/h rated × 37.9 m; motor 90 kW; NPSHa 12.2 m (EQL); baseplate 2.6 × 1.0 m (assumed)
P-104	pump	31.7	50.0	+0.00	2.0 × 0.9 × H 1.0	508 m³/h rated × 9.3 m; motor 22 kW; NPSHa 38.5 m (EQL); baseplate 2.0 × 0.9 m (assumed)
P-102	pump	51.5	53.5	+0.00	2.6 × 1.0 × H 1.2	510 m³/h rated × 36.9 m; motor 90 kW; NPSHa 5.2 m (EQL); baseplate 2.6 × 1.0 m (assumed)
P-103	pump	72.5	50.5	+0.00	1.4 × 0.6 × H 0.8	7 m³/h rated × 27.2 m; motor 2.2 kW; NPSHa 3.7 m (EQL); baseplate 1.4 × 0.6 m (assumed)
K-101	blower	12.0	60.0	+0.00	8.0 × 4.0 × H 4.5	201 × 10³ m³/h at 4.7 kPa; 313 kW shaft at 80 % speed (model) (EQL); casing + motor + VFD-fed, 8 × 4 × 4.5 m (assumed)
VS-301	stack	81.5	56.0	+0.00	D 0.22 × H 27.0	CO2 vent stack DN200 (Relief & vent tab, MEA-PE-RLB-001); outlet +27.0 m chosen by spacing rule R8 (assumed), guyed
CCR	building	13.0	13.0	+0.00	14.0 × 10.0 × H 4.5	Control room / operator shelter (assumed; not on CC-PID-101); door on the east side
SS-101	building	11.0	24.0	+0.00	10.0 × 6.0 × H 4.5	Substation: TR-101 250 kVA, MCC-101, K-101 VFD (Electrical tab MEA-EE-LL-001; position assumed)
AH-101	building	14.0	52.0	+0.00	3.0 × 2.5 × H 3.0	Analyser shelter for the extractive CO2 analysers AI-101 / AI-111 (assumed)

2.2 Spacing rules (typical practice, assumed) and their values on this plan

#	Rule	Limit	On this plan	OK	Basis
R1	Occupied building (control room) to any process equipment	≥ 15 m	15.9 m · CCR / E-105	✓	typical minimum for a non-flammable aqueous amine unit (assumed); hydrocarbon units use 30–60 m. Confirm with a facility siting study.
R2	Control room to the CO2 vent stack VS-301	≥ 30 m	72.2 m · CCR / VS-301	✓	typical (assumed): keep vent outlets away from air intakes of occupied buildings
R3	Substation to any process equipment	≥ 15 m	18.1 m · SS-101 / E-105	✓	typical (assumed): outside the hazardous-area envelope (Electrical tab)
R4	Column to column / structure, shell to edge	≥ 3 m	3.2 m · T-102 / ST-102	✓	typical (assumed): access and erection clearance
R5	Pump baseplate to any other footprint	≥ 1 m	1.2 m · P-102 / ST-102	✓	typical (assumed): maintenance access around the pump
R6	Exchanger bundle pull space (bundle length + 1.5 m) free of equipment	zones clear	all bundle pull zones clear	✓	typical (assumed): TEMA bundle removal
R7	Process equipment to a road edge	≥ 3 m	11.3 m · T-101 / RD-N	✓	typical (assumed): vehicle impact and access
R8	Vent outlet above any platform within 15 m horizontally	≥ 3 m	20.0 m · VS-301 / PF-ST103-2	✓	typical (API 521 style) for atmospheric vents, assumed; CO2 is heavier than air, so a dispersion check is still needed (EQ-L09)
R9	Column centre to the crane road edge	≤ 20 m	14.0 m · T-101 / RD-N	✓	typical (assumed): crane reach for packing and internals removal
R10	Ladder flight between landings	≤ 9 m	9.0 m · PF-T102-1	✓	typical (assumed): caged ladder flight limit, rest platform above
R11	Control room upwind of the process area (prevailing wind)	CCR upwind	cos 1.00 (> 0)	✓	typical: releases drift away from the occupied building
R12	Headroom under pipes crossing a walkway	≥ 2.1 m	2.9 m · DN400-LA-203 over W-2	✓	typical (assumed) clear headroom over walkways; pipes on grade sleepers need a crossover

3 Elevations and levels



Elevations looking north, vertical to scale. Liquid levels at the design point of the model (T-101 sump LIC-109, E-102 lean compartment LIC-208, V-101 LIC-303). Read levels of the field instruments in blue.

3.1 Why the vessels stand where they stand

A pump needs NPSHa ≥ NPSHr + 1 m (typical margin). Rich MEA in the T-101 sump is sub-cooled (vapour pressure well below the column pressure), so a short skirt is enough. Lean MEA leaves the kettle E-102 at its bubble point and the reflux leaves V-101 saturated: for P-102 and P-103 only the static head counts, so E-102 sits on the +5.2 m deck of ST-102 and V-101 on the +3.4 m deck of ST-103. The T-102 bottom is above the kettle so that the kettle fills by gravity, and E-103 drains into V-101 below it.

NPSHa = $(P_s - P_v)/(\rho g) + (z_{LL} - z_{pump}) - h_f$; $h_f = (f \cdot L/D + \Sigma K) \cdot v^2/2g$, f by Swamee–Jain (ε 0.045 mm), ΣK = entrance 0.5 + strainer 1.5 + 0.3 per elbow + gate 0.15 (typical); suction length and elbows from the routed pipe.

Pump	Source	Suction line	P_s kPa a	P_v kPa a	ρ kg/m³	z LL m	L m (layout / HMB)	h_f m	NPSHa m	NPSHr m (assumed)	Margin m	OK
P-101	T-101 sump	DN450-RA-103	107.0	11.6	1,112	+4.12	13.8 / 20	0.135	12.13	4.0	8.13	✓
P-102	E-102 lean compartment	DN500-LA-202	182.4	182.4	1,020	+5.90	13.1 / 10	0.102	5.20	4.0	1.20	✓
P-103	V-101	DN80-RF-306	175.0	175.0	992	+4.32	8.2 / 8	0.035	3.68	2.5	1.18	✓
P-104	T-101 wash draw-off	DN450-WW-106	103.6	11.2	989	+29.70	38.6 / 35	0.187	38.44	3.0	35.44	✓

3.2 Platforms

Platform	On	Level, m	Shape	Ladder from	Instruments read here
PF-T101-1	T-101	+3.0	ring r 2.70–3.90 m	grade	LG-109, LIC-109, TI-108, LSHH-109, LSLL-109
PF-T101-2	T-101	+7.0	ring r 2.70–3.90 m	PF-T101-1	TI-107
PF-T101-3	T-101	+12.0	ring r 2.70–3.90 m	PF-T101-2	TI-106
PF-T101-4	T-101	+17.0	ring r 2.70–3.90 m	PF-T101-3	TI-105
PF-T101-5	T-101	+22.0	ring r 2.70–3.90 m	PF-T101-4	TI-104
PF-T101-6	T-101	+27.0	ring r 2.70–3.90 m	PF-T101-5	TI-103
PF-T101-7	T-101	+32.5	ring r 2.70–3.90 m	PF-T101-6	TIC-110, PDI-102
PF-T102-1	T-102	+9.0	ring r 1.30–2.50 m	grade	LIC-208, LSLL-208, PDI-209
PF-T102-2	T-102	+11.6	ring r 1.30–2.50 m	PF-T102-1	—
PF-T102-3	T-102	+17.0	ring r 1.30–2.50 m	PF-T102-2	—
PF-T102-4	T-102	+21.0	ring r 1.30–2.50 m	PF-T102-3	—
PF-T102-5	T-102	+23.8	ring r 1.30–2.50 m	PF-T102-4	TI-211
PF-ST102	ST-102	+5.2	9.5 × 8.0 m	grade	LG-208, TIC-206, TSHH-206, PI-207
PF-ST103-1	ST-103	+3.4	12.0 × 6.0 m	grade	LIC-303, LG-303, PIC-301, PSHH-301
PF-ST103-2	ST-103	+7.0	12.0 × 10.0 m	PF-ST103-1	TIC-302, TI-312, FI-304, FE-304

3.3 Field instruments: where each one is read

Tag	Kind	Read from	Level, m	Line / item	Description
TI-101	transmitter	grade	+1.5	DUCT-FG-100	Flue gas after the direct-contact cooler (battery limit)
AI-101	analyser	AH-101	+1.5	DUCT-FG-100	CO2 in the flue gas to T-101 (wet basis); design 4.0, range 3–6 (sample from DUCT-FG-100 at x 2.2 m)
FIC-101	transmitter	grade	+1.5	DUCT-FG-100	Flue gas to T-101 → blower K-101 speed (VFD)
PI-101	local gauge	grade	+1.5	DUCT-FG-101	T-101 inlet pressure (blower discharge), model measurement PI101
JI-101	transmitter	SS-101	+1.4	K-101	Blower K-101 shaft power (∝ flow × ΔP) (VFD panel in SS-101)
SI-101	local gauge	SS-101	+1.4	K-101	K-101 speed on the VFD panel (model SI101; VFD display, assumed)
AI-111	analyser	AH-101	+1.5	DUCT-FG-109	CO2 in the treated gas to the stack (sample from DUCT-FG-109 at +37 m)
LG-109	local gauge	PF-T101-1	+4.8	T-101	T-101 sump level gauge glass (reads the sump level, %)
LIC-109	transmitter	PF-T101-1	+4.2	T-101	T-101 sump level → FV-109 rich MEA (P-101, fail closed)
TI-108	transmitter	PF-T101-1	+4.2	T-101	Rich solvent in the T-101 sump
LSHH-109	switch	PF-T101-1	+4.2	T-101	Independent sump high-high switch: 90 % trips I-1 (liquid into the gas inlet)
LSLL-109	switch	PF-T101-1	+4.2	T-101	Sump low-low switch: 5 % stops P-101 (I-3)
TI-103	transmitter	PF-T101-6	+28.2	T-101	T-101 bed 1 top (thermowell at +26.6 m = model cell 1 of 10)
TI-104	transmitter	PF-T101-5	+23.2	T-101	T-101 bed 1 middle (thermowell at +23.2 m = model cell 3 of 10)
TI-105	transmitter	PF-T101-4	+18.2	T-101	T-101 bed 1 bottom (thermowell at +19.9 m = model cell 5 of 10)
TI-106	transmitter	PF-T101-3	+13.2	T-101	T-101 bed 2 middle (thermowell at +14.4 m = model cell 7 of 10)
TI-107	transmitter	PF-T101-2	+8.2	T-101	T-101 bed 2 bottom (thermowell at +9.4 m = model cell 10 of 10)
TIC-110	transmitter	PF-T101-7	+33.7	DUCT-FG-109	Treated gas out of the water wash → TV-110 cooling water to E-105 (fail open)

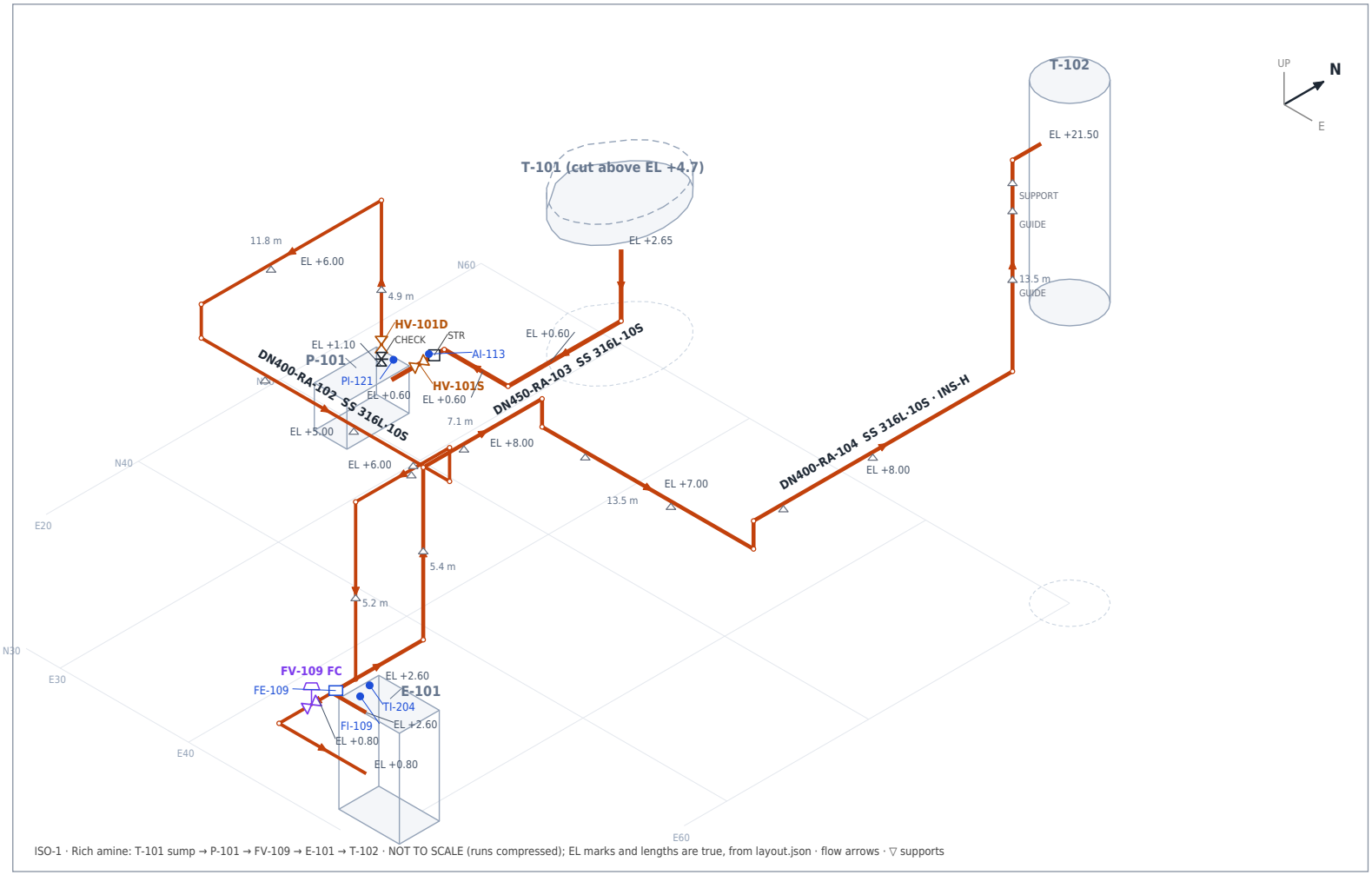
Tag	Kind	Read from	Level, m	Line / item	Description
PDI-102	transmitter	PF-T101-7	+33.7	T-101	T-101 pressure drop (packing, wash, distributors, demister) (taps +7.5 m and +34 m)
PI-121	local gauge	grade	+1.4	DN400-RA-102	P-101 discharge pressure gauge
AI-113	analyser	grade	+1.4	DN400-RA-102	Rich loading, mol CO2 / mol MEA (on-line analyzer) (sample at the P-101 discharge)
PI-124	local gauge	grade	+1.4	DN200-WW-105	P-104 discharge pressure gauge
FIC-201	transmitter	grade	+1.4	DN400-LA-204	Lean MEA to the absorber top (FV-201, fail open)
FE-201	transmitter	grade	+1.0	DN400-LA-204	FIC-201 magnetic flowmeter body (in line)
AI-203	analyser	grade	+1.4	DN400-LA-204	Lean loading, mol CO2 / mol MEA (on-line analyzer) (sample upstream of the T-101 riser)
TI-204	transmitter	grade	+1.5	DN400-RA-104	Rich solvent to T-102 (E-101 cold outlet)
FI-109	transmitter	grade	+1.3	DN400-RA-102	Rich solvent to the stripper
FE-109	transmitter	grade	+0.8	DN400-RA-102	FI-109 magnetic flowmeter body (in line)
TI-205	transmitter	grade	+1.3	DN400-LA-203	Lean solvent after E-101 (to cooler E-104)
AIT-213	analyser	grade	+1.4	DN400-LA-203	MEA concentration (CO2-free basis, density/conductivity) (on LA-203)
TIC-202	transmitter	grade	+1.4	DN400-LA-204	Lean MEA to T-101 after cooler E-104 → TV-202 cooling water (fail open)
TI-222	local gauge	grade	+1.5	DN600-CW-203	E-104 CW return thermometer (model has no E-104 CW outlet variable; v3d computes it, see NOTES)
TI-111	local gauge	grade	+1.5	DN400-CW-111	E-105 CW return thermometer (no model variable)
LG-208	local gauge	PF-ST102	+6.6	E-102	E-102 kettle level gauge (the model level LIC208 is the E-102 inventory: EQ-L05)
TIC-206	transmitter	PF-ST102	+6.6	E-102	Reboiler / T-102 sump temperature → FIC-207 steam set point (cascade)
TSHH-206	switch	PF-ST102	+6.6	E-102	Reboiler high-high 130 °C: MEA thermal degradation, closes the steam (I-2)
PI-207	local gauge	PF-ST102	+6.6	DN400-LS-207	LP steam pressure at the E-102 channel (the model condenses at the header pressure K.Ps; v3d shows K.Ps while FV-207 is open, else falls)
PI-122	local gauge	grade	+1.4	DN400-LA-201	P-102 discharge pressure gauge
FIC-207	transmitter	grade	+1.4	DN400-LS-207	LP steam to reboiler E-102 (FV-207, fail closed)
FE-207	transmitter	grade	+1.0	DN400-LS-207	FIC-207 vortex meter body (in line)
LIC-208	transmitter	PF-T102-1	+10.2	T-102	T-102 sump level → FV-208 make-up water (fail closed)
LSLL-208	switch	PF-T102-1	+10.2	T-102	T-102 sump low-low: 10 % closes the steam (I-2)
PDI-209	transmitter	PF-T102-1	+10.2	T-102	T-102 packing pressure drop
TI-211	transmitter	PF-T102-5	+25.0	T-102	T-102 top temperature (stage 1)
FI-208	transmitter	grade	+1.4	DN25-DW-208	Make-up water to T-102
PI-123	local gauge	grade	+1.3	DN50-RF-303	P-103 discharge pressure gauge
FI-303	transmitter	grade	+1.4	DN50-RF-303	Reflux water to T-102
LIC-303	transmitter	PF-ST103-1	+4.6	V-101	Reflux drum V-101 level → FV-303 reflux to T-102 (P-103, fail closed)
LG-303	local gauge	PF-ST103-1	+4.6	V-101	V-101 magnetic level gauge (reads the drum level, %)
PIC-301	transmitter	PF-ST103-1	+4.6	V-101	Reflux drum V-101 pressure → PV-301 CO2 product to compression (fail open)
PSHH-301	switch	PF-ST103-1	+4.6	V-101	Stripper high-high pressure 250 kPa a closes the steam (I-2); PSV-301 at 350 kPa a
TIC-302	transmitter	PF-ST103-2	+8.2	DN450-OV-305	Condenser E-103 outlet → TV-302 cooling water (fail open)
TI-312	local gauge	PF-ST103-2	+8.4	DN350-CW-303	E-103 CW return thermometer (model x.Tcwout)
FI-304	transmitter	PF-ST103-2	+8.4	DN300-CO-304	CO2 product to compression (dry basis)
FE-304	transmitter	PF-ST103-2	+8.0	DN300-CO-304	FI-304 vortex meter body (in line)
TI-401	transmitter	grade	+1.5	DN750-CW-401	Cooling-water supply

Tag	Kind	Read from	Level, m	Line / item	Description
PI-401	local gauge	grade	+1.5	DN750-CW-401	CW supply header pressure gauge (CW 400 kPa a assumed)
PI-402	local gauge	grade	+1.5	DN50-IA-401	Instrument air header pressure gauge (600 kPa g typical, assumed); falls to 0 when HV-401 is closed

4 Piping isometrics

One sheet per main service. Symbols: bow-tie = valve (orange hand valve, filled = locked closed; purple control valve with actuator and fail position); red triangle = PSV; blue dot = instrument tapping; blue box = in-line flowmeter; ▽ = support (filled = anchor). Not to scale: each axis is compressed (gap d → 1.8·ln(1 + d/0.6) m) so that short runs open up; EL marks and lengths are true. The BOM is counted from the routed geometry and recomputed by the page and the node test.

ISO-1 · Rich amine: T-101 sump → P-101 → FV-109 → E-101 → T-102



ISO-1 · Rich amine: T-101 sump → P-101 → FV-109 → E-101 → T-102 · NOT TO SCALE (runs compressed); EL marks and lengths are true, from layout.json · flow arrows · ▽ supports

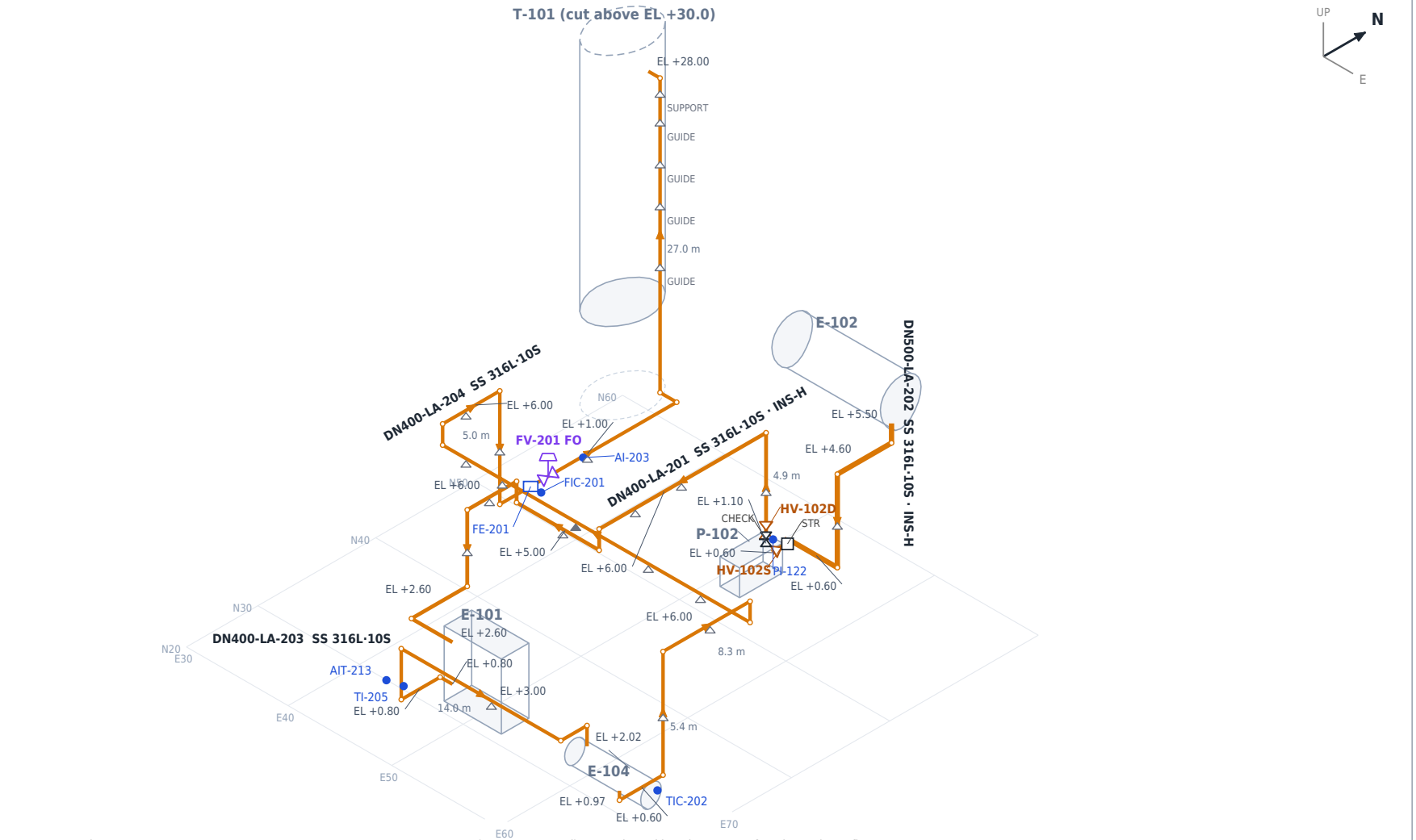
ISO-1 · line table and bill of materials

Line number	Service	DN (sized)	Spec	Ins.	From → to	Operating (HMB)	On P&ID
DN450-RA-103	Rich MEA T-101 sump → P-101 suction (new no.)	450	SS 316L Sch 10S	—	T-101 → P-101	43.6 °C · 122 kPa a · 282.2 t/h	new (EQ-L02)
DN400-RA-102	Rich MEA P-101 → FV-109 → E-101	250 (P&ID 400)	SS 316L Sch 10S	—	P-101 → E-101	43.6 °C · 504 kPa a · 282.2 t/h	yes
DN400-RA-104	Hot rich MEA E-101 → T-102 feed	350 (P&ID 400)	SS 316L Sch 10S	H	E-101 → T-102	109.0 °C · 295 kPa a · 282.2 t/h	yes

Item	Description	Qty	Unit
DN450-RA-103	DN450 · 13.8 m · 3 elbows · 0 tees · 0 supports		
pipe	SS 316L Sch 10S, DN450	13.8	m
elbow 90° LR	DN450	3	ea
flange WN (pairs at in-line items + nozzle ends)	DN450	6	ea
gasket + stud-bolt set	DN450	6	set
gate valve	HV-101S	1	ea
strainer	P-101 suction strainer (typical, untagged)	1	ea
DN400-RA-102	DN250 · 52.5 m · 8 elbows · 0 tees · 7 supports		
pipe	SS 316L Sch 10S, DN250	52.5	m
elbow 90° LR	DN250	8	ea
flange WN (pairs at in-line items + nozzle ends)	DN250	10	ea
gasket + stud-bolt set	DN250	10	set
gate valve	HV-101D	1	ea
globe control valve	FV-109	1	ea
check	P-101 discharge check valve (typical, untagged)	1	ea
flowmeter	FE-109	1	ea
instrument connection (thermowell / tapping / sample)	PI-121, AI-113, FI-109	3	ea
support: rack beam		2	ea
support: riser guide (steel post)		2	ea
support: shoe		3	ea
DN400-RA-104	DN350 · 63.0 m · 9 elbows · 0 tees · 9 supports		
pipe	SS 316L Sch 10S, DN350	63.0	m
elbow 90° LR	DN350	9	ea
flange WN (pairs at in-line items + nozzle ends)	DN350	2	ea
gasket + stud-bolt set	DN350	2	set
instrument connection (thermowell / tapping / sample)	TI-204	1	ea
support: T-post		1	ea
support: guide clip		2	ea
support: load support clip		1	ea
support: rack beam		2	ea
support: riser guide (steel post)		1	ea
support: shoe		2	ea

Item	Description	Qty	Unit
insulation, heat conservation	DN350	63.0	m

ISO-2 · Lean amine: E-102 → P-102 → E-101 → E-104 → FV-201 → T-101



ISO-2 · Lean amine: E-102 → P-102 → E-101 → E-104 → FV-201 → T-101 · NOT TO SCALE (runs compressed); EL marks and lengths are true, from layout.json · flow arrows · ▽ supports

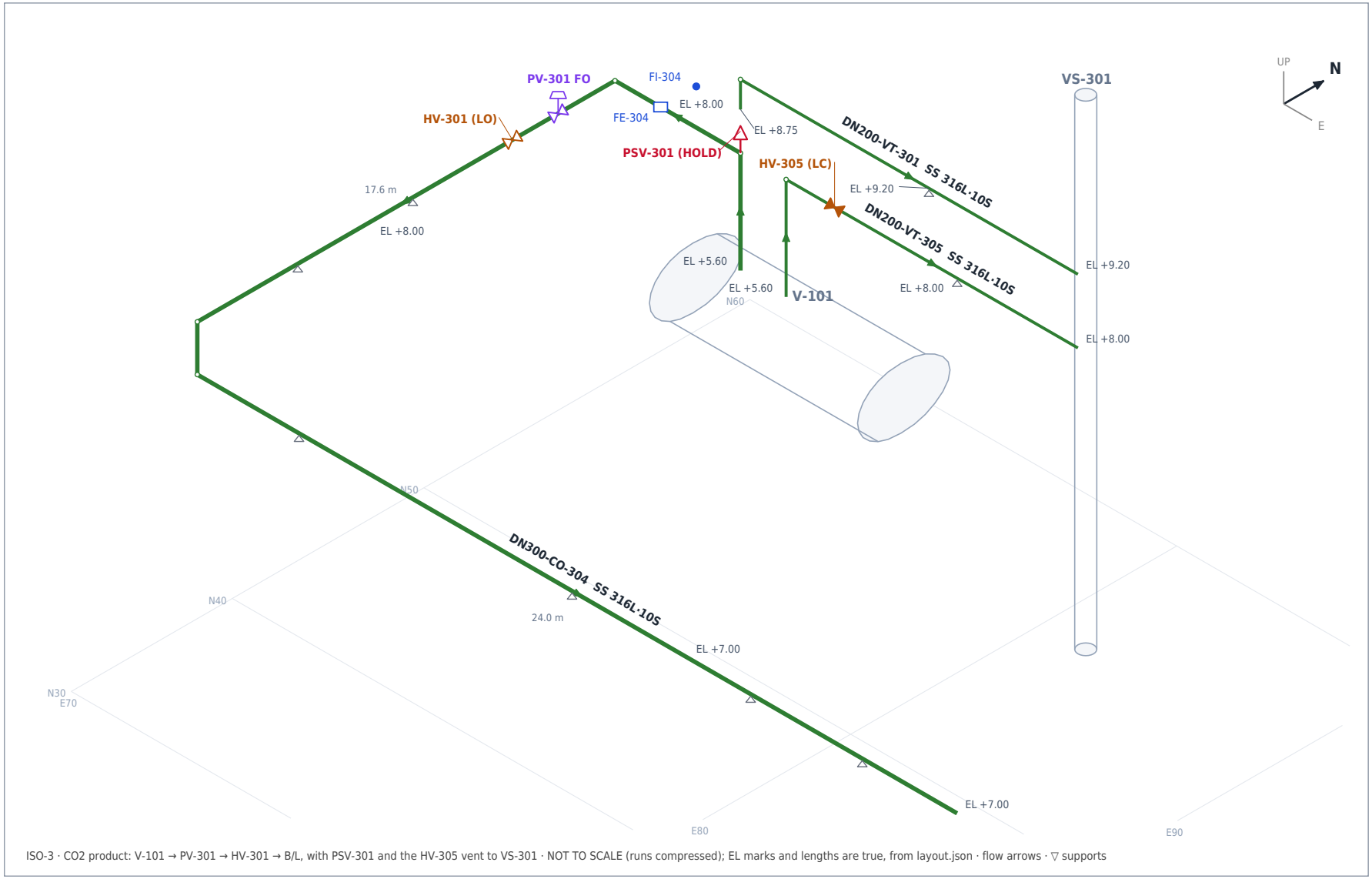
ISO-2 · line table and bill of materials

Line number	Service	DN (sized)	Spec	Ins.	From → to	Operating (HMB)	On P&ID
DN500-LA-202	Lean MEA E-102 → P-102 suction (new no.)	500	SS 316L Sch 10S	H	E-102 → P-102	120.0 °C · 182 kPa a · 269.7 t/h	new (EQ-L02)
DN400-LA-201	Lean MEA P-102 → E-101	300 (P&ID 400)	SS 316L Sch 10S	H	P-102 → E-101	120.0 °C · 495 kPa a · 269.7 t/h	yes
DN400-LA-203	Lean MEA E-101 → E-104	250 (P&ID 400)	SS 316L Sch 10S	—	E-101 → E-104	51.6 °C · 453 kPa a · 269.7 t/h	yes
DN400-LA-204	Lean MEA E-104 → FV-201 → T-101	250 (P&ID 400)	SS 316L Sch 10S	—	E-104 → T-101	40.0 °C · 403 kPa a · 269.7 t/h	yes

Item	Description	Qty	Unit
DN500-LA-202	DN500 · 13.1 m · 4 elbows · 0 tees · 1 supports		
pipe	SS 316L Sch 10S, DN500	13.1	m
elbow 90° LR	DN500	4	ea
flange WN (pairs at in-line items + nozzle ends)	DN500	6	ea
gasket + stud-bolt set	DN500	6	set
gate valve	HV-102S	1	ea
strainer	P-102 suction strainer (typical, untagged)	1	ea
support: riser guide (steel post)		1	ea
insulation, heat conservation	DN500	13.1	m
DN400-LA-201	DN300 · 43.5 m · 8 elbows · 0 tees · 6 supports		
pipe	SS 316L Sch 10S, DN300	43.4	m
elbow 90° LR	DN300	8	ea
flange WN (pairs at in-line items + nozzle ends)	DN300	6	ea
gasket + stud-bolt set	DN300	6	set
gate valve	HV-102D	1	ea
check	P-102 discharge check valve (typical, untagged)	1	ea
instrument connection (thermowell / tapping / sample)	PI-122	1	ea
support: T-post		1	ea
support: rack beam		2	ea
support: riser guide (steel post)		2	ea
support: shoe		1	ea
insulation, heat conservation	DN300	43.4	m
DN400-LA-203	DN250 · 21.5 m · 5 elbows · 0 tees · 1 supports		
pipe	SS 316L Sch 10S, DN250	21.5	m
elbow 90° LR	DN250	5	ea
flange WN (pairs at in-line items + nozzle ends)	DN250	2	ea
gasket + stud-bolt set	DN250	2	set
instrument connection (thermowell / tapping / sample)	TI-205, AIT-213	2	ea
support: T-post		1	ea
DN400-LA-204	DN250 · 102.6 m · 12 elbows · 0 tees · 15 supports		
pipe	SS 316L Sch 10S, DN250	102.6	m

Item	Description	Qty	Unit
elbow 90° LR	DN250	12	ea
flange WN (pairs at in-line items + nozzle ends)	DN250	6	ea
gasket + stud-bolt set	DN250	6	set
globe control valve	FV-201	1	ea
flowmeter	FE-201	1	ea
instrument connection (thermowell / tapping / sample)	FIC-201, AI-203, TIC-202	3	ea
support: anchor on rack bent		1	ea
support: guide clip		4	ea
support: load support clip		1	ea
support: rack beam		2	ea
support: riser guide (steel post)		2	ea
support: shoe		4	ea
support: sleeper at grade		1	ea

ISO-3 · CO2 product: V-101 → PV-301 → HV-301 → B/L, with PSV-301 and the HV-305 vent to VS-301



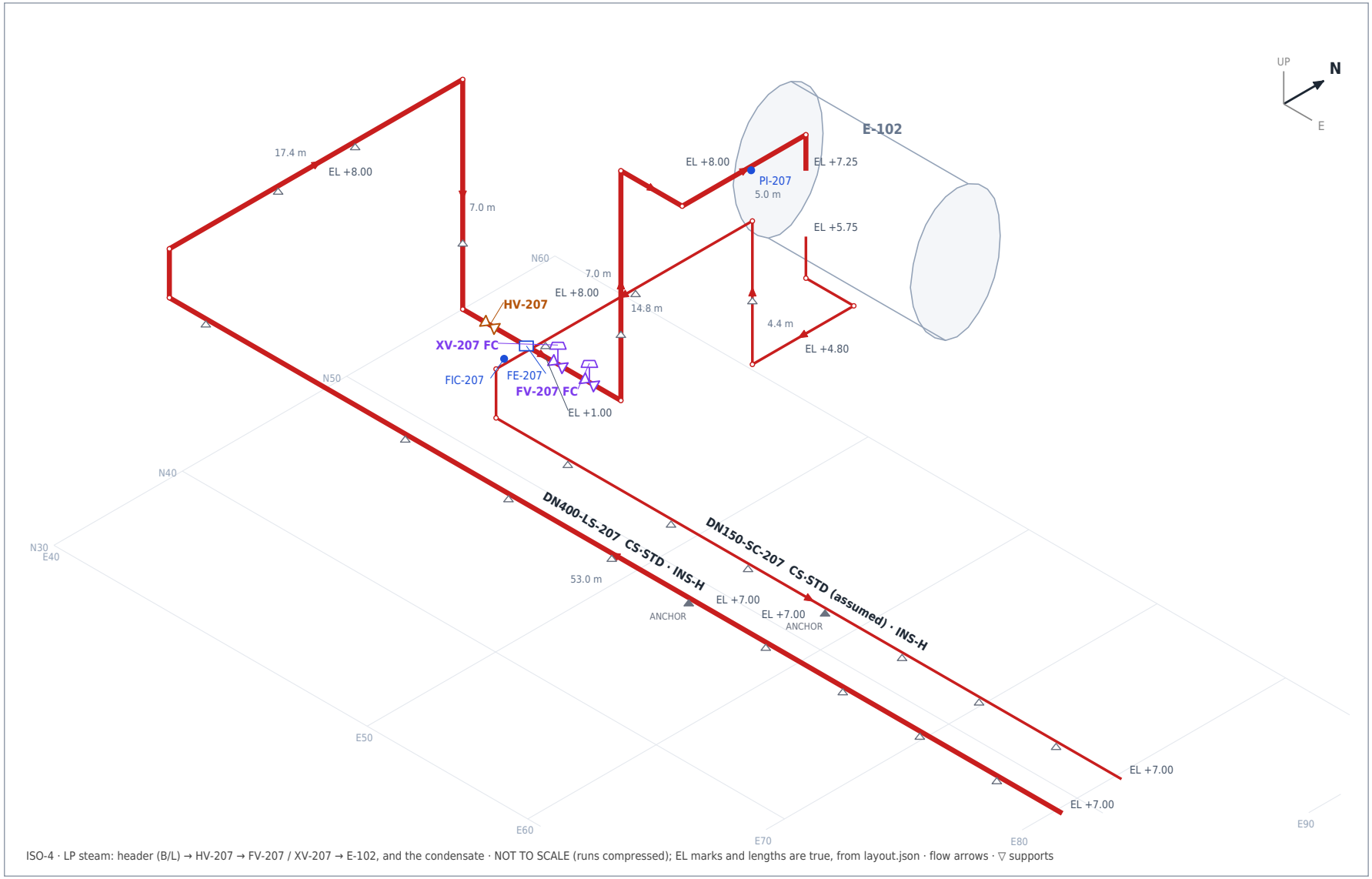
ISO-3 · CO2 product: V-101 → PV-301 → HV-301 → B/L, with PSV-301 and the HV-305 vent to VS-301 · NOT TO SCALE (runs compressed); EL marks and lengths are true, from layout.json · flow arrows · ▽ supports

ISO-3 · line table and bill of materials

Line number	Service	DN (sized)	Spec	Ins.	From → to	Operating (HMB)	On P&ID
DN300-CO-304	CO2 product V-101 → PV-301 → HV-301 → compression (B/L)	400 (P&ID 300)	SS 316L Sch 10S	—	V-101 → B/L	40.0 °C · 175 kPa a · 12.7 t/h	yes
DN200-VT-305	CO2 vent V-101 → HV-305 → VS-301 (new no.)	200	SS 316L Sch 10S	—	V-101 → VS-301	—	new (EQ-L02)
DN200-VT-301	PSV-301 discharge → VS-301 (new no.)	200	SS 316L Sch 10S	—	PSV-301 → VS-301	—	new (EQ-L02)

Item	Description	Qty	Unit
DN300-CO-304	DN400 · 47.4 m · 4 elbows · 1 tees · 6 supports		
pipe	SS 316L Sch 10S, DN400	47.4	m
elbow 90° LR	DN400	4	ea
tee (branch / PSV)	DN400	1	ea
flange WN (pairs at in-line items + nozzle ends)	DN400	8	ea
gasket + stud-bolt set	DN400	8	set
gate valve	HV-301	1	ea
globe control valve	PV-301	1	ea
flowmeter	FE-304	1	ea
instrument connection (thermowell / tapping / sample)	FI-304	1	ea
support: T-post		1	ea
support: rack beam		1	ea
support: shoe		4	ea
DN200-VT-305	DN200 · 10.4 m · 1 elbows · 0 tees · 1 supports		
pipe	SS 316L Sch 10S, DN200	10.4	m
elbow 90° LR	DN200	1	ea
flange WN (pairs at in-line items + nozzle ends)	DN200	4	ea
gasket + stud-bolt set	DN200	4	set
gate valve	HV-305	1	ea
support: beam		1	ea
DN200-VT-301	DN200 · 9.4 m · 1 elbows · 0 tees · 1 supports		
pipe	SS 316L Sch 10S, DN200	9.4	m
elbow 90° LR	DN200	1	ea
flange WN (pairs at in-line items + nozzle ends)	DN200	2	ea
gasket + stud-bolt set	DN200	2	set
support: beam		1	ea

ISO-4 · LP steam: header (B/L) → HV-207 → FV-207 / XV-207 → E-102, and the condensate



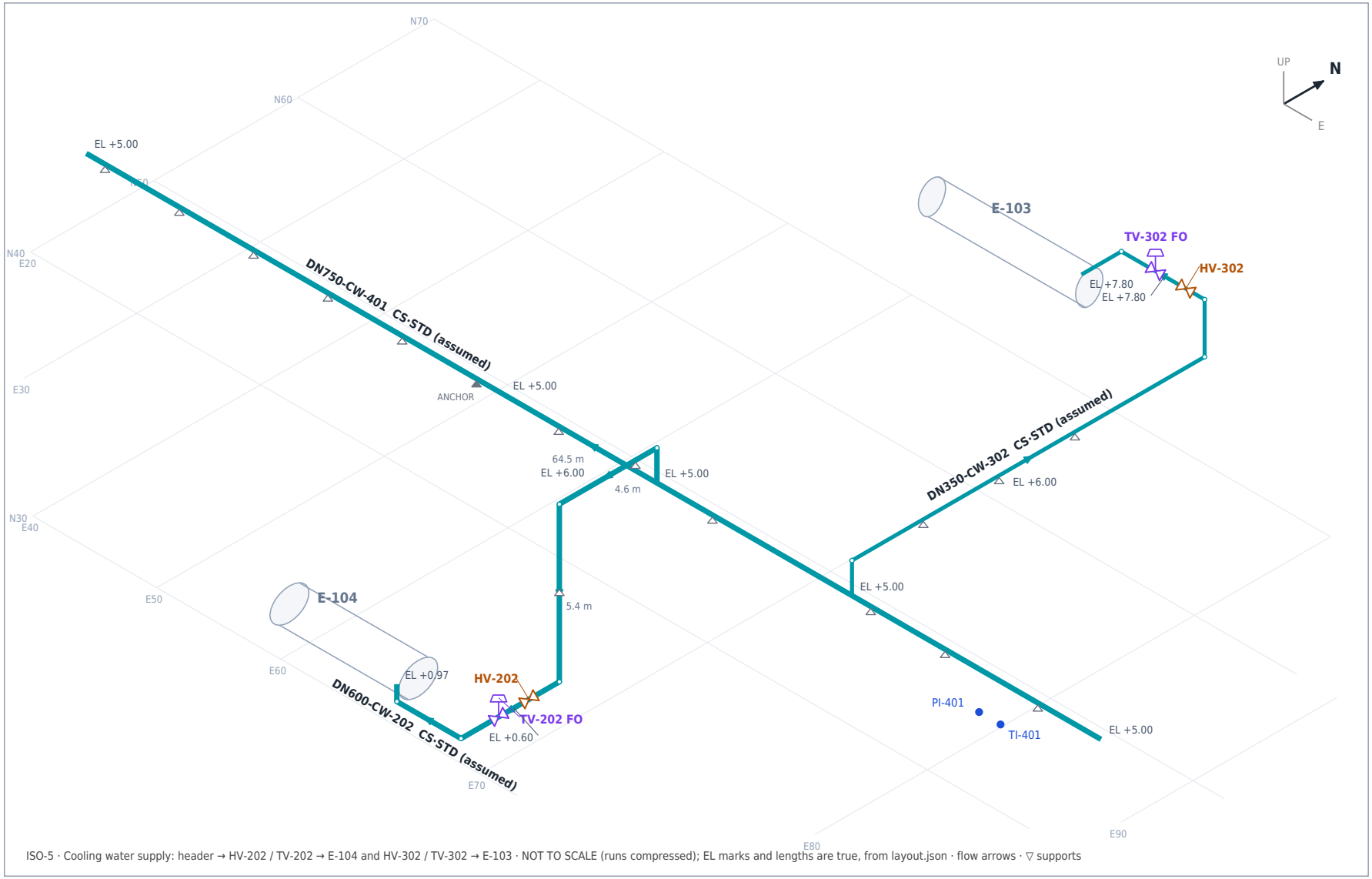
ISO-4 · LP steam: header (B/L) → HV-207 → FV-207 / XV-207 → E-102, and the condensate · NOT TO SCALE (runs compressed); EL marks and lengths are true, from layout.json · flow arrows · ▽ supports

ISO-4 · line table and bill of materials

Line number	Service	DN (sized)	Spec	Ins.	From → to	Operating (HMB)	On P&ID
DN400-LS-207	LP steam header (B/L) → HV-207 → FV-207 / XV-207 → E-102	500 (P&ID 400)	CS STD	H	B/L → E-102	138.9 °C · 350 kPa a · 22.9 t/h	yes
DN150-SC-207	Steam condensate E-102 → B/L (new no.; trap / condensate pot TBC)	150	CS STD (assumed)	H	E-102 → B/L	—	new (EQ-L02)

Item	Description	Qty	Unit
DN400-LS-207	DN500 · 97.8 m · 8 elbows · 0 tees · 13 supports		
pipe	CS STD, DN500	97.8	m
elbow 90° LR	DN500	8	ea
flange WN (pairs at in-line items + nozzle ends)	DN500	9	ea
gasket + stud-bolt set	DN500	9	set
gate valve	HV-207	1	ea
globe control valve	FV-207	1	ea
on-off ball valve	XV-207	1	ea
flowmeter	FE-207	1	ea
instrument connection (thermowell / tapping / sample)	PI-207, FIC-207	2	ea
support: T-post		1	ea
support: anchor on rack bent		1	ea
support: rack beam		1	ea
support: riser guide (steel post)		2	ea
support: shoe		8	ea
insulation, heat conservation	DN500	97.8	m
DN150-SC-207	DN150 · 70.8 m · 6 elbows · 0 tees · 10 supports		
pipe	CS STD (assumed), DN150	70.8	m
elbow 90° LR	DN150	6	ea
flange WN (pairs at in-line items + nozzle ends)	DN150	1	ea
gasket + stud-bolt set	DN150	1	set
support: T-post		1	ea
support: anchor on rack bent		1	ea
support: rack beam		1	ea
support: riser guide (steel post)		1	ea
support: shoe		6	ea
insulation, heat conservation	DN150	70.8	m

ISO-5 · Cooling water supply: header → HV-202 / TV-202 → E-104 and HV-302 / TV-302 → E-103



ISO-5 · Cooling water supply: header → HV-202 / TV-202 → E-104 and HV-302 / TV-302 → E-103 · NOT TO SCALE (runs compressed); EL marks and lengths are true, from layout.json · flow arrows · ▽ supports

ISO-5 · line table and bill of materials

Line number	Service	DN (sized)	Spec	Ins.	From → to	Operating (HMB)	On P&ID
DN750-CW-401	Cooling water supply header (B/L) (new no.)	750	CS STD (assumed)	—	B/L → CAP	—	new (EQ-L02)
DN600-CW-202	CW supply → HV-202 → TV-202 → E-104 (new no.)	600	CS STD (assumed)	—	TEE:CW-401 → E-104	—	new (EQ-L02)
DN350-CW-302	CW supply → HV-302 → TV-302 → E-103 (new no.)	350	CS STD (assumed)	—	TEE:CW-401 → E-103	—	new (EQ-L02)

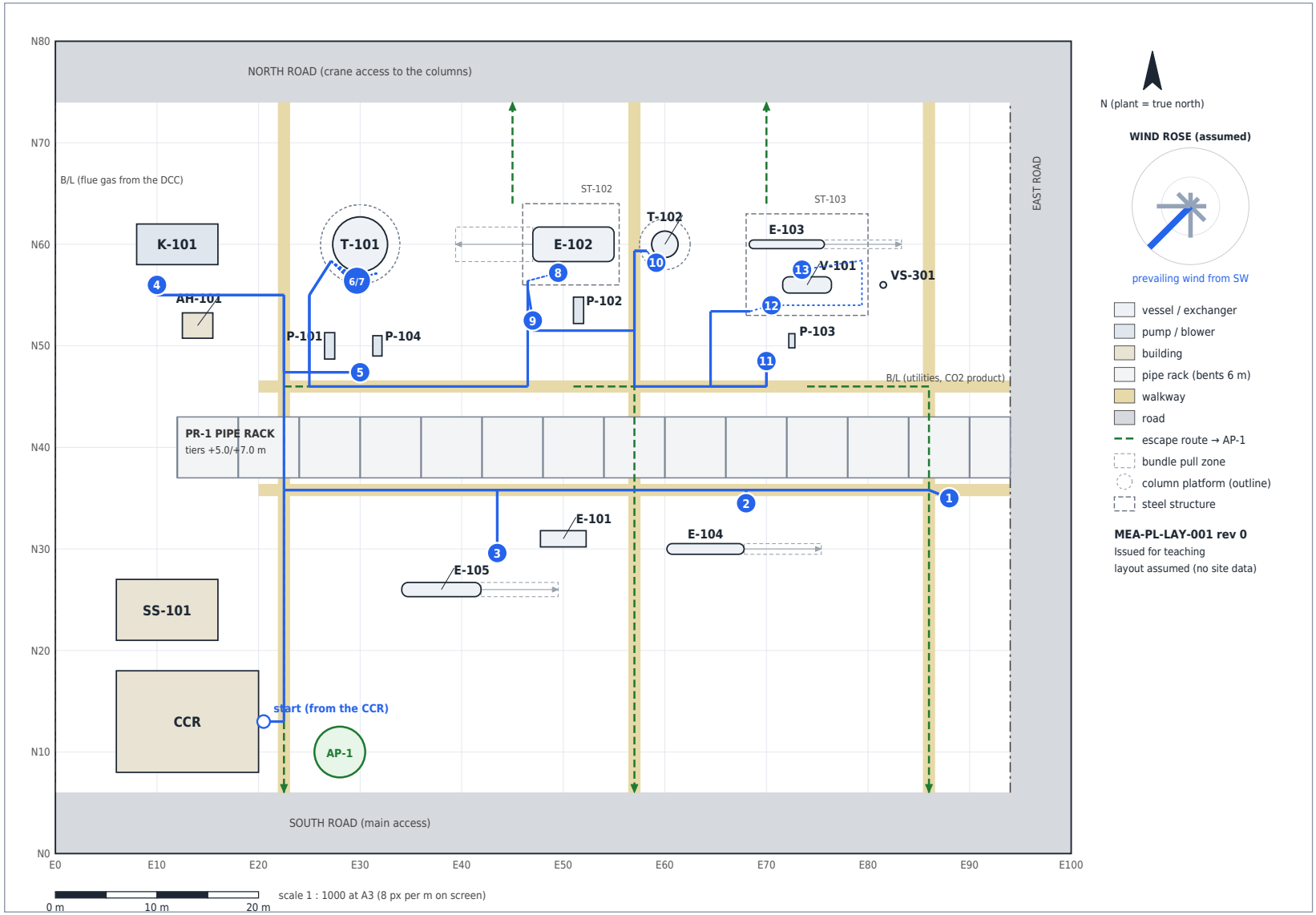
Item	Description	Qty	Unit
DN750-CW-401	DN750 · 64.5 m · 0 elbows · 3 tees · 11 supports		
pipe	CS STD (assumed), DN750	64.5	m
elbow 90° LR	DN750	0	ea
tee (branch / PSV)	DN750	3	ea
flange WN (pairs at in-line items + nozzle ends)	DN750	0	ea
gasket + stud-bolt set	DN750	0	set
instrument connection (thermowell / tapping / sample)	TI-401, PI-401	2	ea
support: anchor on rack bent		1	ea
support: shoe		10	ea
DN600-CW-202	DN600 · 16.2 m · 5 elbows · 0 tees · 2 supports		
pipe	CS STD (assumed), DN600	16.2	m
elbow 90° LR	DN600	5	ea
flange WN (pairs at in-line items + nozzle ends)	DN600	5	ea
gasket + stud-bolt set	DN600	5	set
gate valve	HV-202	1	ea
globe control valve	TV-202	1	ea
support: rack beam		1	ea
support: riser guide (steel post)		1	ea
DN350-CW-302	DN350 · 31.2 m · 4 elbows · 0 tees · 3 supports		
pipe	CS STD (assumed), DN350	31.2	m
elbow 90° LR	DN350	4	ea
flange WN (pairs at in-line items + nozzle ends)	DN350	5	ea
gasket + stud-bolt set	DN350	5	set
gate valve	HV-302	1	ea
globe control valve	TV-302	1	ea
support: T-post		1	ea
support: beam		1	ea
support: rack beam		1	ea

5 From layout to the 3-D model and the operator round

The tool `tools/mk_layout.py` writes the plan, elevations and routing into `layout.json`; Visual plant 3D draws the same file. The node test `tools/test_layout.js` rejects it if a footprint overlaps, a pipe passes through equipment or another pipe, a pipe end misses its nozzle, the route walks through an item, or a tag of the equipment list or HV_DEF is missing.

Key	What it holds	Count
`equipment`	boxes / cylinders: tag, type, centre x y, base z0, D H L W, orient, nozzles (id, service, x y z)	17
`platforms`	ring or rect, level z, ladder from, ladder points	15
`racks`	plan points, tier levels, width, bent spacing	1
`pipes`	line number, service, DN, OD, spec, polyline [x,y,z], from / to, supports	34
`instruments`	tag, x y z, read from, kind, model measurement key, range	58
`handValves`	HV_DEF tag, x y z, line, normal position, lock	14
`route`	stations: n, name, eye point, look-at, via points, items	13

5.1 The operator round on the plot plan



Solid blue = walking at grade; dotted = on a platform or ladder. Station numbers as in the table.

#	Station	Stands at	Items checked
1	East battery limit: cooling water and instrument air	E 88.0 N 35.0 · floor +0.0 m	TI-401, PI-401, **HV-401**, PI-402
2	Lean amine cooler E-104	E 68.0 N 34.5 · floor +0.0 m	**HV-202**, TV-202, TIC-202, TI-222
3	Lean/rich exchanger E-101 and wash-water cooler E-105	E 43.5 N 29.6 · floor +0.0 m	FI-109, FV-109, TI-204, TI-205, AIT-213, **HV-110**, TV-110, TI-111
4	Flue gas blower K-101 and analyser shelter AH-101	E 10.0 N 56.0 · floor +0.0 m	XV-101, TI-101, FIC-101, PI-101, AI-101, AI-111
5	T-101 base: rich pump P-101, wash pump P-104, lean valve FV-201	E 30.0 N 47.4 · floor +0.0 m	**HV-101S**, **HV-101D**, PI-121, AI-113, **HV-104D**, PI-124, FIC-201, FV-201, AI-203
6	T-101 sump platform PF-T101-1 (+3.0 m)	E 29.7 N 56.4 · floor +3.0 m	LG-109, LIC-109, TI-108, LSHH-109, LSLL-109
7	T-101 top platform PF-T101-7 (+32.5 m)	E 29.7 N 56.4 · floor +32.5 m	TIC-110, PDI-102
8	Reboiler E-102 deck PF-ST102 (+5.2 m)	E 49.5 N 57.2 · floor +5.2 m	TIC-206, TSHH-206, LG-208, PI-207
9	Lean pump P-102 and the LP steam valve station	E 47.0 N 52.5 · floor +0.0 m	**HV-102S**, **HV-102D**, PI-122, **HV-207**, FIC-207, XV-207, FV-207
10	T-102 sump platform PF-T102-1 (+9.0 m)	E 59.2 N 58.2 · floor +9.0 m	LIC-208, LSLL-208, PDI-209
11	Reflux pump P-103 and the make-up water station	E 70.0 N 48.5 · floor +0.0 m	**HV-103D**, PI-123, FI-303, FV-303, **HV-208**, FV-208, FI-208
12	Reflux drum V-101 deck PF-ST103-1 (+3.4 m)	E 70.5 N 54.0 · floor +3.4 m	LG-303, LIC-303, PIC-301, PSHH-301
13	Condenser E-103 and CO2 product deck PF-ST103-2 (+7.0 m)	E 73.5 N 57.5 · floor +7.0 m	TV-302, **HV-302**, TIC-302, TI-312, FI-304, PV-301, **HV-301**, PSV-301, **HV-305**

6 Engineering queries

EQ	Subject	Documents	Finding	Suggested resolution
EQ-L01	Hand valves not on the P&ID	CC-PID-101 rev A vs this layout and the model (HV_DEF)	The 14 hand valves of the model (HV-101S, HV-101D, HV-102S, HV-102D, HV-103D, HV-104D, HV-110, HV-202, HV-302, HV-207, HV-208, HV-401, HV-301, HV-305) are placed here on their lines for the field round; CC-PID-101 rev A does not draw them.	Add the block valves, their lock (LO/LC) symbols and car-seal list to CC-PID-101 rev B.
EQ-L02	Line numbers that are not on the P&ID	Layout / isometrics vs CC-PID-101 rev A	New numbers used here: DUCT-FG-100, DUCT-FG-109, DN450-RA-103, DN450-LA-205, DN500-LA-202, DN450-WW-106, DN450-OV-305, DN80-RF-306, DN200-VT-305, DN200-VT-301, DN150-SC-207, DN50-IA-401, DN750-CW-401, DN750-CW-402, DN600-CW-202, DN600-CW-203, DN350-CW-302, DN350-CW-303, DN400-CW-110, DN400-CW-111.	Issue the numbers in the line list (MEA-PE-HMB-001) and on CC-PID-101 rev B; the isometrics keep them until then.
EQ-L03	Line size in the P&ID number differs from the sized line	CC-PID-101 rev A vs MEA-PE-HMB-001 part 4 (EQ-P01)	The line numbers keep the rev A DN prefix; pipes, BOM and the 3-D model use the sized DN: DN400-RA-102 → DN250; DN400-RA-104 → DN350; DN500-VR-302 → DN700; DN400-LA-201 → DN300; DN400-LA-203 → DN250; DN400-LA-204 → DN250; DN200-WW-105 → DN300; DN300-CO-304 → DN400; DN400-LS-207 → DN500.	Renumber at rev B with the sized DN.
EQ-L04	Cooling-water flows are very large for the duties	Model valve capacities K.Wcwl / Wcwc / Wcww vs typical CW practice	At design the model sends 509 kg/s to E-104 (TV-202 51 %), 221 kg/s to E-103 and 249 kg/s to E-105: rises of about 1.5, 4.4 and 5.5 K. Sized at 2.5 m/s this needs DN600, DN350, DN400 branches and a DN750 header. A typical CW rise is 8–10 K.	Re-rate the CW side of E-104 (and E-103/E-105) for an 8–10 K rise, reduce the TV capacities in the model, and re-size the CW lines.
EQ-L05	Which vessel LIC-208 measures	MEA-IN-INL-001 / CC-PID-101 ("T-102 sump level") vs the model (levR = E-102 kettle inventory) and MEA-PE-HMB-001 (P-102 NPSH from the E-102 level)	The layout puts LT-208 and LS-208 on the T-102 sump as the P&ID shows, and a gauge LG-208 on E-102 that reads the model level.	Decide: level in the kettle lean compartment (behind the weir) is the one that protects P-102; move LIC-208 / LSLL-208 there on rev B.
EQ-L06	Local gauges added for the field round	Layout vs MEA-IN-INL-001	Added (assumed): PI-101, LG-109, PI-121, PI-124, TI-222, TI-111, LG-208, PI-207, PI-122, PI-123, LG-303, TI-312, PI-401, PI-402.	Add them to the instrument list (local, no I/O) and to the P&ID.
EQ-L07	E-102 condensate system not shown	CC-PID-101 rev A, MEA-PE-HMB-001	Line SC-207, a condensate pot or trap and its level control are not on the P&ID; the layout routes SC-207 to the B/L.	Add the condensate pot (level control on the condensate) and the line number.
EQ-L08	Suction lengths from the layout vs the pump hydraulics	MEA-PE-HMB-001 part 5 (equivalent lengths 20/10/8/35 m) vs this layout	The page recomputes NPSHa with the routed suction lengths (part 3); see the margins there.	Update the HMB pump sheets with the routed lengths at the next issue.
EQ-L09	CO2 vent and relief discharge location	MEA-PE-RLB-001 (VS-301 DN200) vs this layout	VS-301 outlet is set at +27.0 m by rule R8; CO2 is denser than air, so the 3 m / 15 m rule alone does not prove safe dispersion to the T-102 top platform and the ST-103 decks.	Dispersion study of the HV-305 and PSV-301 cases (Consequence tab method), then fix the outlet height.
EQ-L10	Control room distance basis	Spacing rule R1 (assumed 15 m)	The 15 m value assumes a non-flammable solvent; CO2 from VS-301 and hot MEA leaks are the occupied-building hazards.	Facility siting study; consider a blast/toxic-rated shelter or move the CCR further from E-105.
EQ-L11	XV-101 position	CC-PID-101 rev A (XV-101 upstream of K-101 at the B/L) vs I-1 intent	The damper is placed at the battery limit as drawn; it then isolates the blower suction, not the absorber inlet.	Confirm the position; a second damper at the T-101 inlet may be needed for maintenance isolation.

7 Checks against the model and the other documents

36 of 36 checks ✓. The same function (laChecks in lay.core.js) runs in the tab; this table is its result in node at the model design point.

#	Check	Expected	This layout / model	OK
C1	Every EQL item (MEA-ME-EQL-001) is on the plot plan	13 items	13 of 13	✓
C2	No two equipment footprints overlap	0 overlaps	0	✓
R1	Spacing: Occupied building (control room) to any process equipment	≥ 15 m	15.9 m · CCR / E-105	✓
R2	Spacing: Control room to the CO2 vent stack VS-301	≥ 30 m	72.2 m · CCR / VS-301	✓
R3	Spacing: Substation to any process equipment	≥ 15 m	18.1 m · SS-101 / E-105	✓
R4	Spacing: Column to column / structure, shell to edge	≥ 3 m	3.2 m · T-102 / ST-102	✓
R5	Spacing: Pump baseplate to any other footprint	≥ 1 m	1.2 m · P-102 / ST-102	✓
R6	Spacing: Exchanger bundle pull space (bundle length + 1.5 m) free of equipment	zones clear	all bundle pull zones clear	✓
R7	Spacing: Process equipment to a road edge	≥ 3 m	11.3 m · T-101 / RD-N	✓
R8	Spacing: Vent outlet above any platform within 15 m horizontally	≥ 3 m	20.0 m · VS-301 / PF-ST103-2	✓
R9	Spacing: Column centre to the crane road edge	≤ 20 m	14.0 m · T-101 / RD-N	✓
R10	Spacing: Ladder flight between landings	≤ 9 m	9.0 m · PF-T102-1	✓
R11	Spacing: Control room upwind of the process area (prevailing wind)	CCR upwind	cos 1.00 (> 0)	✓
R12	Spacing: Headroom under pipes crossing a walkway	≥ 2.1 m	2.9 m · DN400-LA-203 over W-2	✓
C3	Every hand valve of the model (HV_DEF) is on the layout	14 valves	14 of 14	✓
C4	Hand-valve normal position and lock (LO/LC) = HV_DEF	all match	all match	✓
C5	Every hand valve sits on its pipe and is a stop on the operator round	on line and on the round	14 on line · 14 on the round	✓
C6-1	ISO-1: every line number is on CC-PID-101 or listed in EQ-L02	DN450-RA-103, DN400-RA-102, DN400-RA-104	2 on the P&ID · 1 new (DN450-RA-103) in EQ-L02	✓
C6-2	ISO-2: every line number is on CC-PID-101 or listed in EQ-L02	DN500-LA-202, DN400-LA-201, DN400-LA-203, DN400-LA-204	3 on the P&ID · 1 new (DN500-LA-202) in EQ-L02	✓
C6-3	ISO-3: every line number is on CC-PID-101 or listed in EQ-L02	DN300-CO-304, DN200-VT-305, DN200-VT-301	1 on the P&ID · 2 new (DN200-VT-305, DN200-VT-301) in EQ-L02	✓
C6-4	ISO-4: every line number is on CC-PID-101 or listed in EQ-L02	DN400-LS-207, DN150-SC-207	1 on the P&ID · 1 new (DN150-SC-207) in EQ-L02	✓
C6-5	ISO-5: every line number is on CC-PID-101 or listed in EQ-L02	DN750-CW-401, DN600-CW-202, DN350-CW-302	0 on the P&ID · 3 new (DN750-CW-401, DN600-CW-202, DN350-CW-302) in EQ-L02	✓
C7	Every pipe run in layout.json carries a P&ID or EQ-listed line number	33 line numbers	all found	✓
C8	Control valves on the isometrics are P&ID tags	11 valves	all on CC-PID-101	✓
C9	No pipe-to-pipe clash (OD/2 + OD/2 + 25 mm)	0	0	✓
C10	Pipes clear of equipment they do not connect to	0	0	✓
C11	Walking route clear of footprints and of pipes below 2.1 m headroom	0	0	✓
C12	Every route point stands on grade or on a platform (eye height 1.6 m)	13 stations	all on grade / platform	✓
C13	Bill of materials recomputed from the pipe runs = the issued BOM	33 lines	all equal	✓
C14-101	P-101: NPSHa at the low-low level − NPSHr ≥ 1.0 m (model design point)	≥ 1.0 m (NPSHr 4 m assumed)	8.13 m (NPSHa 12.13 m)	✓
C14-102	P-102: NPSHa at the low-low level − NPSHr ≥ 1.0 m (model design point)	≥ 1.0 m (NPSHr 4 m assumed)	1.20 m (NPSHa 5.20 m)	✓
C14-103	P-103: NPSHa at the low-low level − NPSHr ≥ 1.0 m (model design point)	≥ 1.0 m (NPSHr 2.5 m assumed)	1.18 m (NPSHa 3.68 m)	✓
C14-104	P-104: NPSHa at the low-low level − NPSHr ≥ 1.0 m (model design point)	≥ 1.0 m (NPSHr 3 m assumed)	35.44 m (NPSHa 38.44 m)	✓
C15	Column and kettle diameters = model (K.Da, K.Ds, K.Dr)	5.4 / 2.6 / 3.4 m	5.4 / 2.6 / 3.4 m	✓

#	Check	Expected	This layout / model	OK
C16	Instruments read on the round map to a model measurement	36 with a meas key	all in sim.meas	✓
C17	Every item named at a route station exists in layout.json	72 items, 13 stations	all exist	✓

Appendix A · Hand valves (HV_DEF) on the layout

Tag	Line	E, m	N, m	EL, m	Normal	Lock	Station	Description
HV-101S	DN450-RA-103	27.0	52.2	+0.60	open	—	5	P-101 rich pump suction block valve · added for the field round (EQ-L01)
HV-101D	DN400-RA-102	27.0	51.0	+1.60	open	—	5	P-101 rich pump discharge block valve · added for the field round (EQ-L01)
HV-102S	DN500-LA-202	51.5	55.3	+0.60	open	—	9	P-102 lean pump suction block valve · added for the field round (EQ-L01)
HV-102D	DN400-LA-201	51.5	54.5	+1.60	open	—	9	P-102 lean pump discharge block valve · added for the field round (EQ-L01)
HV-103D	DN50-RF-303	72.5	50.9	+1.40	open	—	11	P-103 reflux pump discharge block valve · added for the field round (EQ-L01)
HV-104D	DN200-WW-105	31.7	50.7	+1.60	open	—	5	P-104 wash water pump discharge block valve · added for the field round (EQ-L01)
HV-110	DN400-CW-110	42.2	28.0	+0.50	open	—	3	Cooling water supply to wash water cooler E-105 · added for the field round (EQ-L01)
HV-202	DN600-CW-202	69.0	32.0	+0.60	open	—	2	Cooling water supply to lean cooler E-104 · added for the field round (EQ-L01)
HV-302	DN350-CW-302	77.0	62.0	+7.80	open	—	13	Cooling water supply to condenser E-103 · added for the field round (EQ-L01)
HV-207	DN400-LS-207	41.8	55.0	+1.00	open	—	9	LP steam block valve upstream of FV-207 · added for the field round (EQ-L01)
HV-208	DN25-DW-208	66.0	47.8	+1.00	open	—	11	Demin make-up water block valve upstream of FV-208 · added for the field round (EQ-L01)
HV-401	DN50-IA-401	92.5	34.0	+1.20	open	—	1	Instrument air header isolation valve · added for the field round (EQ-L01)
HV-301	DN300-CO-304	70.0	54.0	+8.00	open	LO	13	CO2 product block valve downstream of PV-301 (car-sealed open: relief path) · added for the field round (EQ-L01)
HV-305	DN200-VT-305	74.5	56.0	+8.00	closed	LC	13	CO2 vent to atmosphere at V-101 (locked closed) · added for the field round (EQ-L01)